

Fig. 3: Actual-size PCB pattern of the optoreflexive sensor

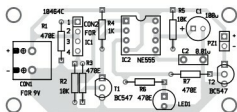


Fig. 4: Component layout of the PCB

Construction and testing

An actual-size, single-side PCB pattern for the optoreflexive sensor is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the PCB.

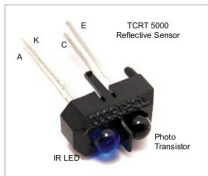


Fig. 5: Pin configuration of TCRT5000 sensor

Use connector CON1 for battery connection and connector CON2 (shown in Fig. 4) for connecting the TCRT5000 sensor module.

Use a 9-volt PP3 battery or a suitable DC power supply. Even though the phototransistor has a filter casing, sunlight triggers the circuit. So test the circuit inside a room where only artificial light is present. Output from the optoreflexive sensor can be interfaced with a microcontroller or similar circuits.

Refer Fig. 5 for pin configuration of TCRT5000 sensor before using it in the circuit. Use a 4-wire cable for interconnection between the sensor and the PCB. Connect pins K, C, A and E of the sensor module to pins 1, 2, 3 and 4 of CON2, respectively. Fix the sensor module on the front panel of the cabinet or enclosure. **END**



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